

MOPLR073

High gradient accelerating structures for carbon therapy linac*

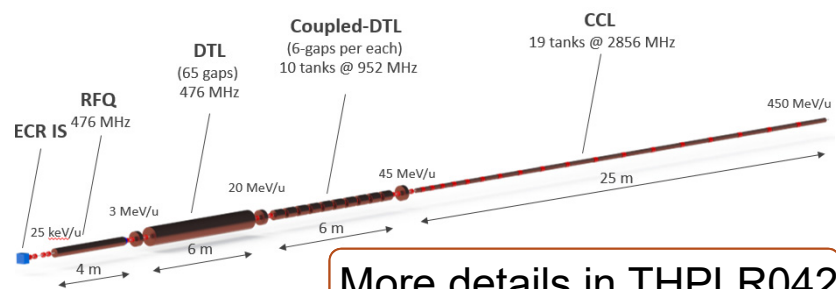
Sergey V Kutsaev

09/26/2016

Oral poster at LINAC' 16 Conference

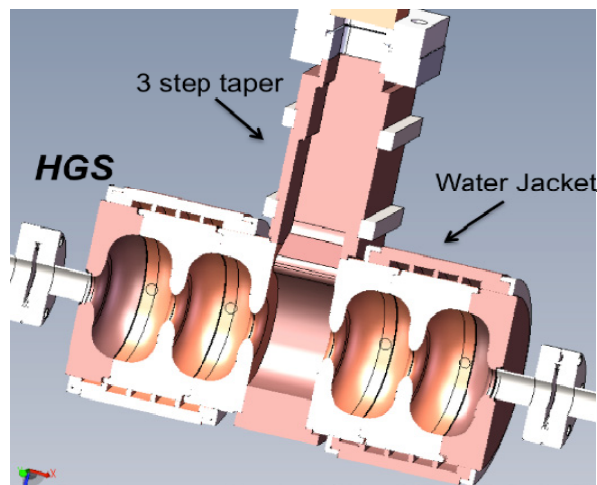
ACCIL: Hadron therapy linac

- An Advanced Compact Carbon high gradient Ion Linac (ACCIL) is being developed by collaboration of Argonne National Laboratory and RadiaBeam Systems



- 1 GV accelerating voltage in a 40m length
- 90% of energy gain in 60% of length!
- ~35 MV/m real-estate gradients and 50 MV/m accelerating gradients

- High power tests of RadiaBeam high-gradient structure* (HGS) with $\beta=1$ structure was done at ANL



- Demonstrated 52 MV/m
- at 16.7 MW power
- 2 μ s pulse length
- at 30 Hz rep rate,
- sustained for 24 hours.

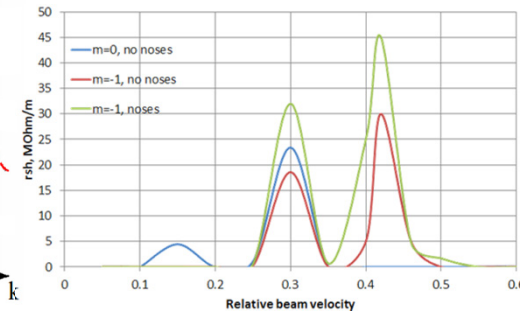
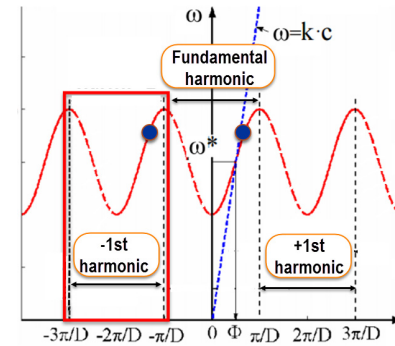
Further attempts to increase the peak fields in the cavity by increasing pulse width resulted in an internal failure of the cavity

* L. Faillace et al., "Fabrication and Initial Tests of an Ultra-High Gradient Compact S-Band (HGS) Accelerating Structure", Proceedings of IPAC2012, New Orleans, Louisiana, USA, p. 3392

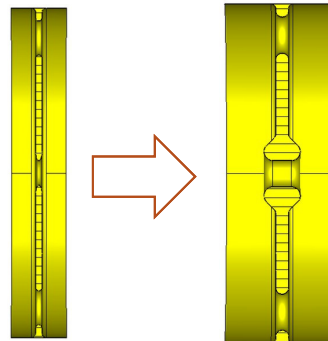
Low-beta Structures

- We used backward travelling wave (BTW) structure for medium-beta design and higher spatial harmonics structure for low-beta

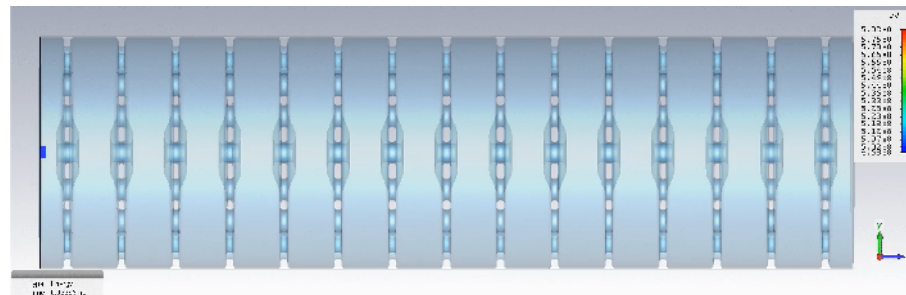
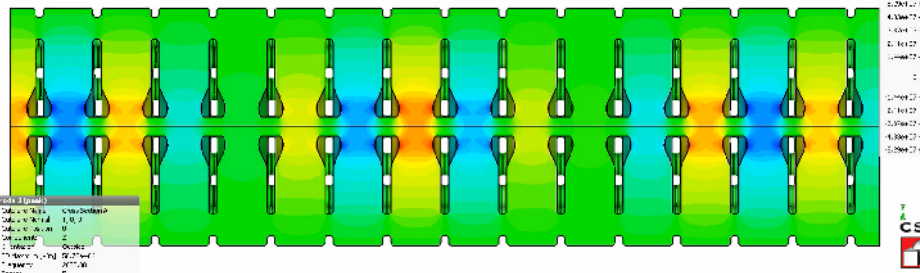
β	0.43	0.5	0.6	0.7
Mode	$2\pi/3$	$5\pi/6$	$5\pi/6$	$5\pi/6$
Section length, m	0.6	0.79	1.1	1.16
$R_{sh,eff}$, M Ω /m	36	56	57	58
Filling time, μ s	1.0	1.0	1.6	1.72
Required power, MW	49	40	54	54



- HGS is optimal for $\beta > 0.7$,
- BTW for $\beta = 0.5..0.7$,
- HHC for $\beta = 0.3-0.4$



Harmonic number	0	-1	-1
Optimal mode	$2\pi/3$	$5\pi/6$	$5\pi/6$
Noses	No	No	Yes
Shunt impedance, M Ω /m	22.0	18.6	31.7
Peak surface E-field, MV/m	92.5	130	156.5
Pulse heating (@1.5 μ s), K	24	33	28



* S. Benedetti, A. Degiovanni, A. Grudiev et al., Proceedings of LINAC2014, Geneva, Switzerland.

Thank You!

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 - Luigi Faillace,
 - Sergey Kutsaev
 - Evgeny Savin.
- Argonne Team:
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 - Brahim Mustapha,
 - Peter Ostroumov,
 - Alexander Plastun.
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- Visit us at MOPLR073 and THPLR042 posters!